Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **OpEx**
b. Team leader name: **Srihitha Pulapa**
c. Problem Statement: **Manufacturing · Intelligent Operations & Industrial Efficiency**

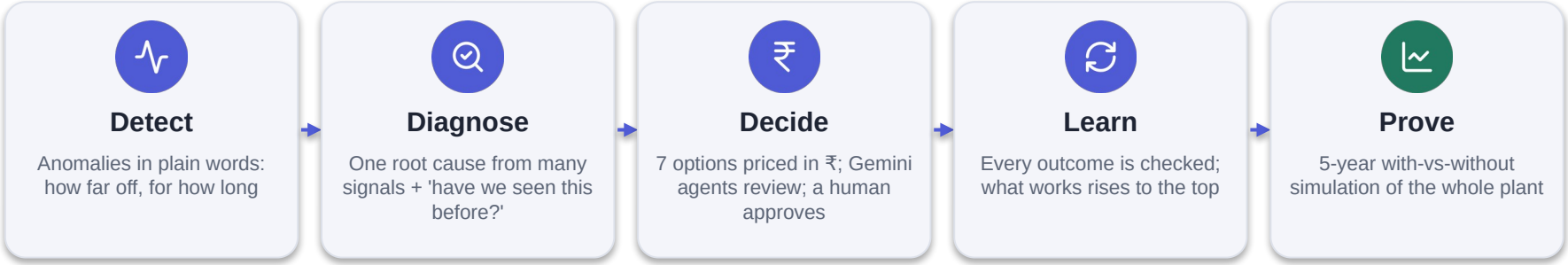
OpEx

From machine signals to money-backed decisions

Brief about the idea

Spot trouble early, price every fix in rupees, let a person decide

India's ₹7.6 lakh-crore auto-component industry (ACMA FY26) still runs most CNC shops reactively: **plants sit at 60–75% OEE against a world-class 85%**, alarms flood screens, and a surprise spindle failure can cost a 24-machine plant ₹3–4 lakh in one go.



−74.5%
breakdowns over 5 simulated years

₹7.11 Cr
net benefit after paying for OpEx

12.1×
return · pays back in month 3

10,012 h
machine-hours of breakdown avoided

Simulated results for a 24-machine plant, calibrated on 4 real industrial datasets (AI4I 2020, NASA C-MAPSS, Azure PdM, Zenodo line OEE). Calculation on slide 'How we calculated the 5-year numbers'.

Opportunities

Not another dashboard: a decision engine with proof of value

How is it different?

- Combines signals into ONE cause instead of 50 separate alarms
- Prices every option in ₹ (repair, downtime, lost output, scrap, energy, risk), not just 'risk: high'
- Admits doubt: bad sensor data visibly lowers confidence; a faulty sensor is called a sensor fault
- Learns from each decision's real outcome in this plant

How does it solve the problem?

- Warns hours to weeks ahead, so parts are ready and swaps happen in the Sunday window
- Engineers see the trade-off and approve in one screen
- Operators ask in their own language, by voice, with gloves on
- Managers see ₹ impact, investigations and the 5-year business case

USP

- Gemini incident room: Diagnostician, Planner and Safety-officer agents, grounded on computed facts and SOPs
- 'Rewind the factory': replays a real breakdown with OpEx watching
- Honest numbers: misses, false alarms and people ignoring advice are all in the simulation
- Accessible by design: 6 Indian languages, read-aloud, high contrast, dyslexia-friendly font

Capability	Typical CMMS / PdM dashboard	OpEx
Root cause across machine, quality and energy	Single-signal thresholds	Signature matching + past incidents + chain of impact
What to do and what it costs	Work order only	7 priced options, multi-objective ranking, safety rule
Proof of value before buying	Vendor case studies	5-year calibrated with-vs-without simulation of your plant

List of features offered by the solution

Eight capabilities, one flow: detect, diagnose, decide, learn, prove

Live factory health

OEE (A×P×Q), 5 health scores, ranked 'worry about first' list, floor map

Anomaly → root cause

'Vibration 148% above normal for 52 min', chain of impact, sensor-fault detection

Failure & quality prediction

8-h failure probability, 10–90% window, remaining life per component, defect forecast

Priced options

Stop, slow 10%, continue, shift load, partial fix, replace, monitor: each in ₹

AI incident room + approval

3 Gemini agents review; approve / reject / modify by role; full audit log

Ask OpEx copilot

Function-calling Gemini, voice in/out, 6 Indian languages, part-photo check

Closed-loop learning

Outcome checked after 2 plant hours; '9 of 11 times' feeds ranking; AI investigations

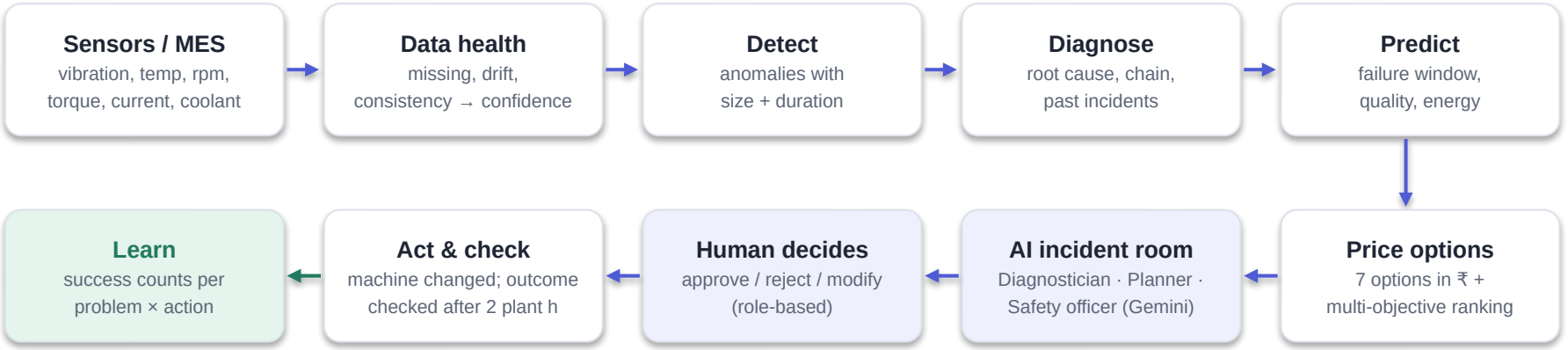
5-year simulation + rewind

Choose replay length; live with/without counters; ROI; CSV downloads

Also: data-health checks (missing, drift, calibration), maintenance plan + spare parts, Pub/Sub alerts, energy standby advice, CO_2 tracking, sensor CSV upload checker, factory-memory search, 4 roles with server-side permissions.

Process flow diagram or Use-case diagram

Every recommendation passes a person and is checked afterwards



↺ Learn feeds real success rates back into 'Price options' for the next decision

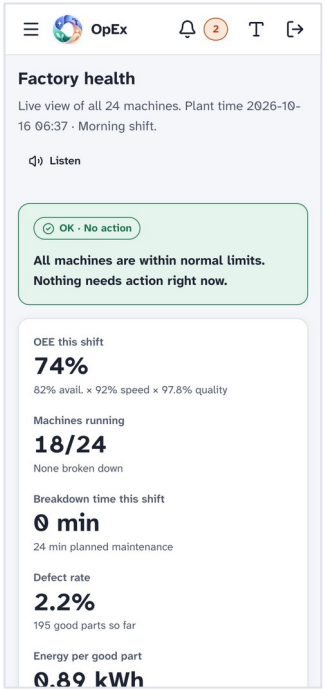
Operator
asks by voice, acknowledges alerts, approves small fixes

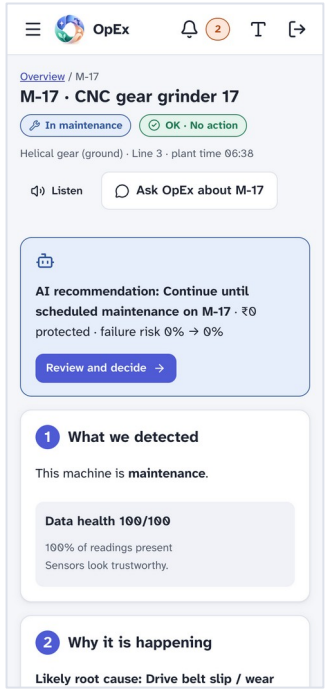
Maintenance engineer
approves stops and part swaps, writes investigations

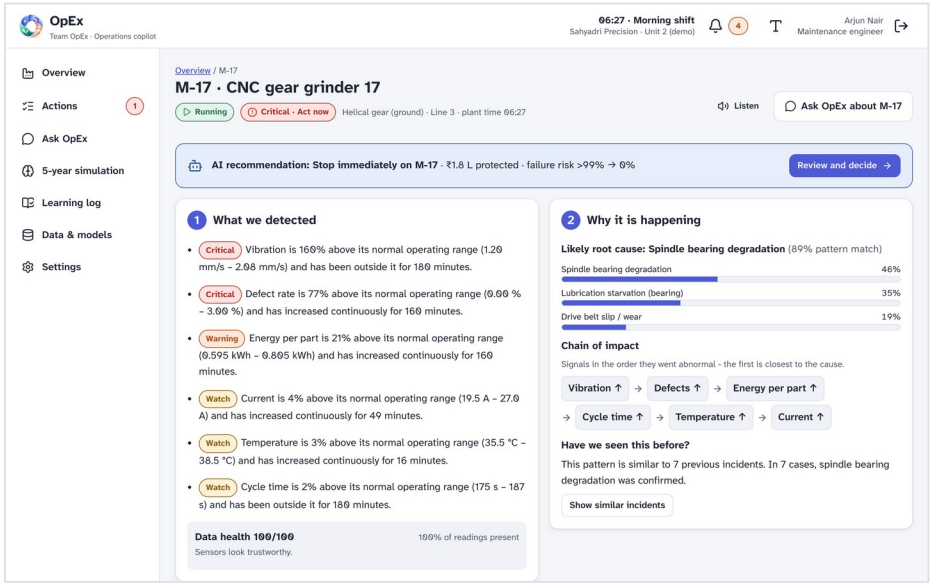
Plant manager
sets priorities and ₹ assumptions, runs the 5-year case

Wireframes/Mock diagrams of the proposed solution (optional)

Designed for the shop floor: phone at the machine, desktop in the office



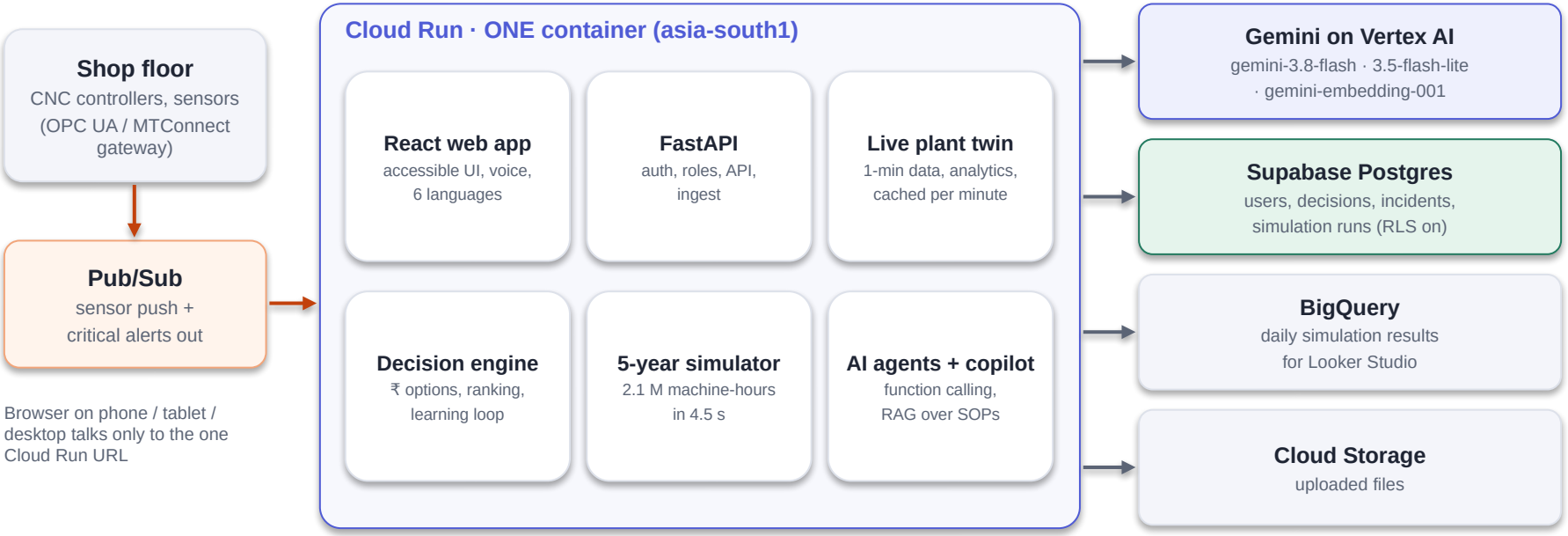




- Numbered steps on every machine: 1 What we detected · 2 Why · 3 What next · 4 What should we do
- Status = icon + word + colour (ISA-101 style: grey when normal, colour only for trouble)

Architecture diagram of the proposed solution

One Cloud Run service + Gemini on Vertex AI + Supabase Postgres



Offline pipeline: AI4I 2020 · NASA C-MAPSS · Azure PdM · Zenodo line OEE · NIST Nestor → clean → train 3 models → calibrate simulator → build factory memory (793 jobs, 194 incidents)

Technologies to be used in the solution

Google Cloud where it adds value, and nothing used just for show

Technology	What OpEx uses it for
Gemini on Vertex AI (gemini-3.8-flash, 3.5-flash-lite)	3-agent incident room, function-calling copilot (6 tools), investigations, part-photo check, 6 Indian languages
gemini-embedding-001 (768-d)	Semantic search in factory memory (SOPs, manuals, 194 past incidents) + keyword hybrid
Cloud Run	Single service: web app + API + live twin + simulator + agents
BigQuery	Daily with/without simulation results for long-term analysis and Looker Studio
Pub/Sub	Critical alerts out to SMS / WhatsApp / CMMS; sensor readings in (push to /api/ingest)
Cloud Storage	Uploaded sensor files and documents
Supabase Postgres	Users, roles, decisions, outcomes, incidents, runs; Row Level Security on
Python · FastAPI · scikit-learn	Analytics, 3 ML models, 5-year simulator (numpy, shared random draws)
React · TypeScript · Recharts	Accessible web app: WCAG 2.2 AA, Atkinson Hyperlegible, voice, read-aloud

Deliberately not used: Dataflow and Vertex AI Training. At 24 machines and 1-minute data, training takes 30 s in the container. Dataflow (Pub/Sub → BigQuery) is the path past ~10k readings/second.

Estimated implementation cost (optional)

₹16.8 L to start, ₹8.4 L a year, for a 24-machine plant

Item	How it is calculated	Amount
Sensor retrofit	₹45,000 vibration + current + temperature kit × 24 machines	₹10.8 L
Setup and training	Integration, commissioning, operator training (one time)	₹6.0 L
Cloud (per year)	Cloud Run + Gemini + Supabase Pro + BigQuery/GCS (below)	₹2.4 L
Support (per year)	Model upkeep, help desk, updates	₹6.0 L
5-year total	₹16.8 L once + ₹8.4 L × 5 years	₹58.8 L

Cloud cost per month (estimate)

- Gemini: ~300 agent/copilot calls a day × 6k tokens = 54 M tokens/month → 40 M in × $0.75 + 14 M out × $3.75 ≈ $82 ≈ ₹7k
- Cloud Run 2 vCPU / 2 GiB, mostly idle: ≈ ₹4–6k
- Supabase Pro $25 ≈ ₹2.1k · BigQuery + Storage: free tier
- **Total ≈ ₹15–20k/month ≈ ₹2.4 L/year**

₹7.69 Cr
5-year benefit (simulated)

₹58.8 L
5-year cost of OpEx

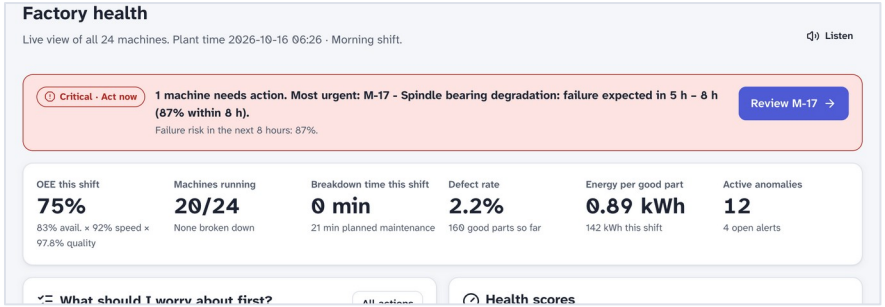
12.1×
net return on cost

month 3
payback

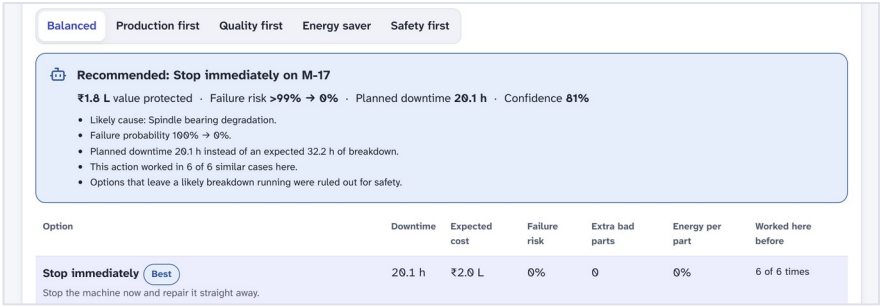
Gemini prices: Google AI pricing page, Oct 2026 (gemini-3.8-flash $0.75 / $3.75 per M tokens until 31 Dec 2026). USD at ₹84.

Snapshots of the prototype

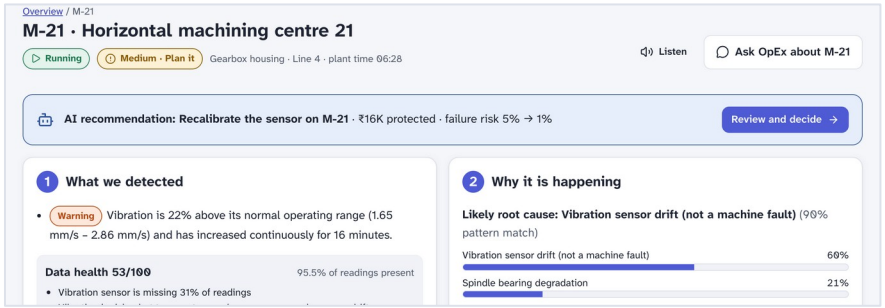
Working prototype: four real screens from the running app



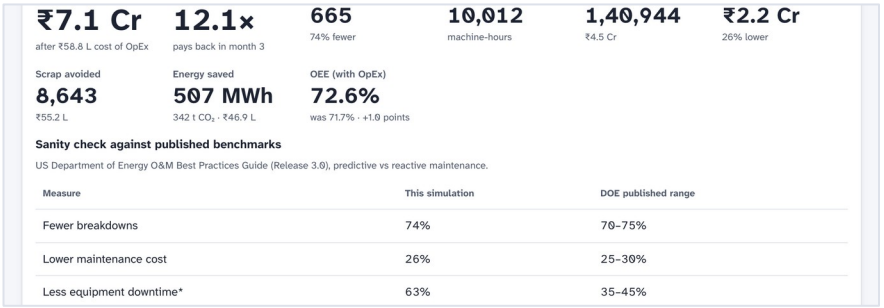
Factory health and 'worry about first'



Options priced in ₹ + human approval



Faulty sensor caught: recalibrate, not a false breakdown alarm

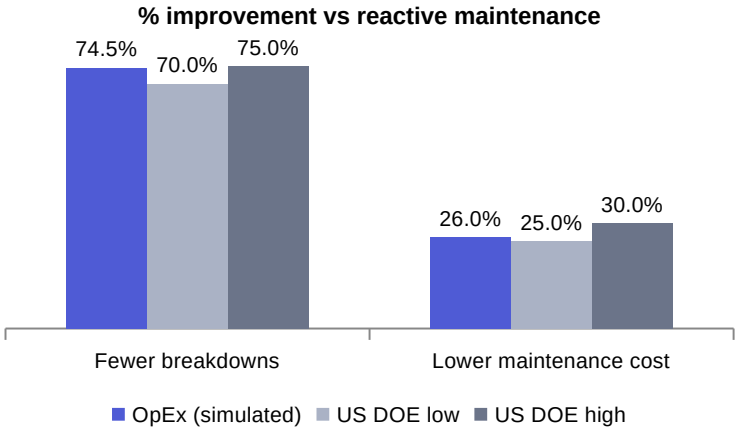


5-year results vs published benchmarks

Prototype Performance report/Benchmarking

Scored on unseen data; results land inside published benchmarks

Model (held-out test)	Data	Score
Failure-mode classifier	AI4I 2020 · 2,000 test rows	ROC AUC 0.996 · P 94% / R 89%
Remaining-life model	NASA C-MAPSS FD001 · 100 test engines	RMSE 13.0 cycles · NASA score 300
Remaining-life 10–90% band	same	covers 82% of engines
Component precursor model	Azure PdM · Sep–Dec held out	caught 94–100% of failures 24–32 h ahead
Live overview (24 machines)	app benchmark	0.18 s warm · 5-yr simulation 4.5 s

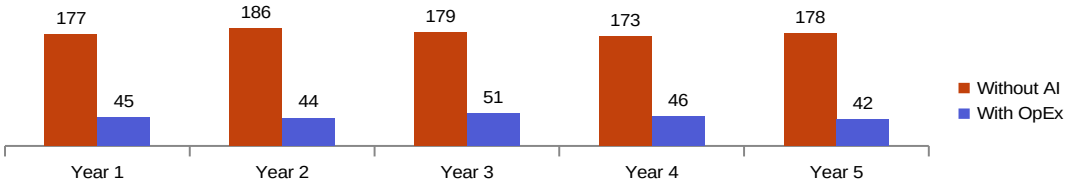


- **Honest by design:** the simulation counts model misses (sensor coverage 90%), people ignoring advice (15%), 1,364 false alarms over 5 years (each a 1 h inspection stop) and parts stock-outs.
- **Equipment downtime** −63% (DOE 35–45% counts all stops; our material and set-up stops, 17.5% of planned time, stay identical in both worlds).
- **Tested:** 7 automated tests (simulator determinism and benefit, live scenario, roles and decision flow); SQL verified on Postgres 18 with Supabase-style roles.

How we calculated the 5-year numbers

Same plant, same luck, two policies: where the ₹7.1 Cr comes from

Benefit line	Formula	Result
Production protected	140,944 extra good parts × contribution ₹175–760 per part	₹4.45 Cr
Maintenance saved	parts + labour (×1.5 overtime after breakdowns) + secondary damage + inspections	₹2.22 Cr
Scrap avoided	8,643 fewer bad parts × scrap cost ₹380–1,450	₹55.2 L
Energy saved	507 MWh × ₹9.25/kWh (MSEDCL HT FY27) · 342 t CO_2 at 0.675 t/MWh (CEA v22)	₹46.9 L
Total benefit	sum of the four lines above	₹7.69 Cr
Net benefit / ROI	₹7.69 Cr − ₹58.8 L cost = ₹7.11 Cr · ÷ cost	12.1×



Calibrated inputs

- Failures/machine-yr: bearing 1.79 · drive 2.59 · electrical 1.92 · coolant 1.31 (Azure PdM)
- Wear curve damage(u) = (e^3.75u − 1)/(e^3.75 − 1); life spread Weibull k 4.41 (NASA C-MAPSS)
- Prevented = recall 0.94–1.00 × sensor coverage 0.90 × follow-advice 0.85 ≈ 72–76%
- Warning timing error N(0.04, 0.20) of remaining life (C-MAPSS out-of-fold)
- Minor stops 8.9% + set-ups 8.6%, repair-time σ 0.82 (Zenodo real line)
- 6,750 planned h/yr (3 shifts × 7.5 h × 300 days)

Breakdowns per year 893 → 228 over 5 years. OEE 71.7% → 72.6%: modest by design, because material and set-up losses are untouched.

Additional Details/Future Development (if any)

Built to scale from one plant to India's 1,100+ ACMA members

Next 3 months

- Pilot on 6 machines with OPC UA / MTConnect gateway
- Re-fit health estimators per machine
- Shift-handover summaries by Gemini

6–12 months

- Multi-plant view and benchmarking
- Dataflow streaming into BigQuery
- Per-machine fine-tuning on Vertex AI with 6 months of data

Later

- Line-camera visual inspection with Gemini
- Spare-parts auto-ordering
- Energy tariff-aware scheduling (peak hours +25%)

Community benefit

- 1.5 M people work directly in Indian auto components (ACMA): fewer emergency night repairs, safer work with LOTO steps built in
- Works in Hindi, Telugu, Tamil, Kannada, Marathi; read-aloud and voice for low-literacy and low-vision workers
- 342 t CO_2 avoided per 24-machine plant over 5 years (simulated)

Feasibility and scalability

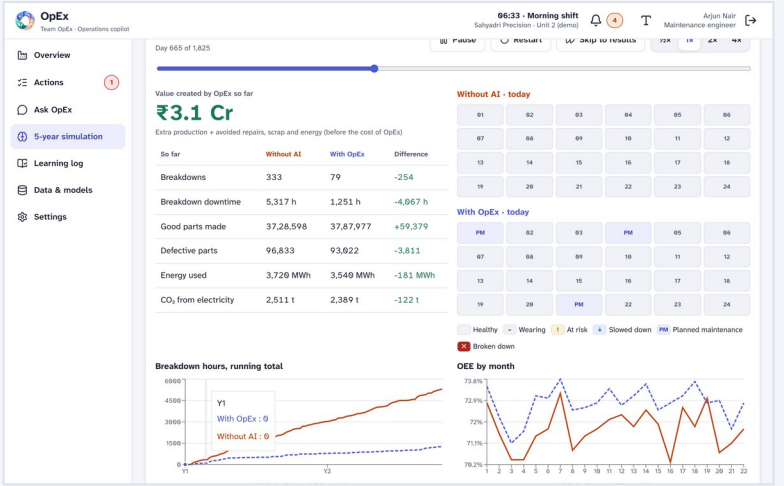
- Runs today: one container, zero keys needed to try, one script to deploy
- Sensor-agnostic models (health-index features) transfer to any machine type
- Scale-out path: real ingest replaces the twin → Cloud Run instances → pgvector at 100k documents

Provide links to your:

- GitHub Public Repository: github.com/<your-team>/opex (add after pushing)
- Demo Video Link (3 Minutes): <add YouTube / Drive link> (script in HOW_TO_RUN.md)
- Final Product Link: <add Cloud Run URL> (./deploy/deploy_cloud_run.sh prints it)

Try it in 60 seconds

- Click 'Explore as a guest' on the sign-in page, or sign in as engineer@opex.demo (password OpEx-demo-2026)
- Open M-17 → see the diagnosis → compare priced options → AI incident room → Approve
- 5-year simulation → choose replay length → Simulate → Skip to results → Rewind the factory
- Settings → Display: try high contrast, extra-large text and Lexend



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Build for the world.

Thank you

